

List of publication and products

Prof. Dr. Pierros Ntelis

June 9, 2026

Via, a 14 *h*-index, a 16 *i10*-index, and total citations: 4002 ([Google Scholar 2026](#)), my most important work is represented in the following papers:

Accepted

- [1] P. Yovkochev and **Ntelis, P.**, 2026. Evolution and Mass Decay of a Gaussian Pulse in Time-Fractional Diffusion: An L1-Implicit Scheme Analysis. *Turanian Journal*, vol. 2, no. 1, pp. 020103-1–020103-7. [URL](#), Contribution: Supervision, and co-design of analysis. (Citations: 0)
- [2] **Ntelis, P.** 2025. Advancing Tensor Theories. Symmetry: Mathematics: Advances in Topology and Algebraic Geometry. [URL](#) (Citations: 2) ★
- [3] **Ntelis, P.** 2025. Advanced manifold-metric pairs. Mathematics. [ArXiv](#), [URL](#) (Citations: 1) ★
- [4] **Ntelis, P.** & J. L. Said, 2025. Analytical *poly* Λ CDM dynamics. PDU.[URL](#) (Citations: 1) ★
- [5] E.D. Valentino, J. L. Said, ..., **Ntelis, P.** ..., et al. 2025. The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics. Contribution: Writing on extended theories of gravity, scientific review, and suggestions for further development. PDU [URL](#) (Citations: 472)
- [6] **Ntelis, P.** 2022. A (D_τ, D_x) -manifold with N -correlators of N_t -objects. IJGMMP, arXiv e-prints arXiv:2209.07472 [URL](#), (Citations: 0)
- [7] **Ntelis, P.** & J. L. Said 2025. Exploring ϕ CDM model dynamics. EPJC, arXiv:2502.03486v1 [URL](#), (Citations: 4)
- [8] **Ntelis, P.** & J. L. Said 2025. Simple $\phi\Lambda$ CDM dynamics. IJGMMP, arXiv:2502.03486v2 [URL](#), (Citations: 3)
- [9] S.Casas, ..., **Ntelis, P.**, and 128 colleagues 2025. Euclid: Constraints on f(R) cosmologies from the spectroscopic and photometric primary probes. A&A, [URL](#) (Citations: 27) Contribution: Scientific review, software development, and contribution on discussions for further research.
- [10] Euclid Collaboration, Y. Mellier and 1137 colleagues 2024. Euclid. I. Overview of the Euclid mission. A&A, arXiv:2405.13491 [URL](#), (Citations: 785) Contribution: Scientific review and contribution on discussions for further research.
- [11] **Ntelis, P.** & Morris, A. 2023. Functors of actions. Foundations of Physics, arXiv:2010.06707 [URL](#), (Citations: 24) ★
- [12] Göksel Karaçaylı, N., ... **Ntelis, P.**, and 10 colleagues 2022. Optimal 1D Ly α Forest Power Spectrum Estimation – II. KODIAQ, SQUAD & XQ-100. MNRAS [URL](#), (Citations: 58), Contribution: Scientific review and contribution on discussions for further research and software development.
- [13] Hamaus, N., ... **Ntelis, P.**, and 128 colleagues 2022. Euclid: Forecasts from redshift-space distortions and the Alcock-Paczynski test with cosmic voids. A&A arXiv e-prints [URL](#), (Citations: 95), Contribution: Scientific review and contribution on discussions for further research and software development.
- [14] Ilić, S., ... **Ntelis, P.**, and 207 colleagues 2021. *Euclid* preparation: XV. Forecasting cosmological constraints for the *Euclid* and CMB joint analysis. A&A arXiv e-prints [URL](#), (Citations: 104), Contribution: Scientific review and contribution on discussions for further research. ★
- [15] Alam, S., ... **Ntelis, P.**, and 33 colleagues 2020. Towards testing the theory of gravity with DESI: summary statistics, model predictions and future simulation requirements. JCAP [URL](#),

- (Citations: 75), Contribution: Scientific review and contribution on discussions for further research and software development. ★
- [16] Tutusaus, I., ... **Ntelis, P.**, and 89 colleagues 2020. Euclid: The importance of galaxy clustering and weak lensing cross-correlations within the photometric Euclid survey. *Astronomy and Astrophysics* 643, [URL](#), (Citations: 92), Contribution: Scientific review and contribution on discussions for further research.
- [17] **Ntelis, P.**, A. Ealet, S. Escoffier, J.-C. Hamilton, A. J. Hawken, J.-M. Le Goff, J. Rich, and A. Tilquin 2018. The scale of cosmic homogeneity as a standard ruler. *Journal of Cosmology and Astroparticle Physics* 2018, [URL](#), (Citations: 28)
- [18] Blanton, M. R., ... **Ntelis, P.**, and 361 colleagues 2017. Sloan Digital Sky Survey IV: Mapping the Milky Way, Nearby Galaxies, and the Distant Universe. *The Astronomical Journal* 154, [URL](#), (Citations: 2122), Contribution: Scientific review and contribution on discussions for further research. ★
- [19] **Ntelis, P.**, and 22 colleagues 2017. Exploring cosmic homogeneity with the BOSS DR12 galaxy sample. *Journal of Cosmology and Astroparticle Physics* 2017, [URL](#), (Citations: 113) ★
- [20] Laurent, P., ... **Ntelis, P.**, and 17 colleagues 2016. A $14 h^{-3} \text{ Gpc}^3$ study of cosmic homogeneity using BOSS DR12 quasar sample. *Journal of Cosmology and Astroparticle Physics* 2016, [URL](#), (Citations: 99), Contribution: Scientific review and contribution on discussions for further research and software development.

Under review

- [1] **Ntelis, P.**, et al. 2019. Cosmological constraints from cosmic homogeneity. (Cites: 4) [URL](#)
- [2] **Ntelis, P.** 2025. A probabilistic and expanding Universe. [URL](#), (Cites: 2)
- [3] **Ntelis, P.** 2023. Advancing category and tensor theories. [URL](#), (Cites: 2)
- [4] **Ntelis, P.** 2025. CosmoFATs. [URL](#), (Cites: 2)
- [5] **Ntelis, P.** 2025. CosmoFATs Dynamics. [URL](#) (Cites: 0)
- [6] **Ntelis, P.** 2025. A cosmic Schiff's conjecture test. [URL](#), , [DOI](#), (Cites: 0)
- [7] **Ntelis, P.** 2025. A universal foundational theory. [URL](#), (Cites: 0)
- [8] **Ntelis, P.** 2026. · A dot. [URL](#), [DOI](#), (Cites: 0)
- [9] **Ntelis, P.** 2026. Analytical sub-poly Λ CDM through cosmic obs. [URL](#), [DOI](#), (Cites: 0)
- [11] **Ntelis, P.** 2026. A foundational analysis of local functional calculus [URL](#), [DOI](#), (Cites: 0)
- [11] **Ntelis, P.**, B.J. Ahmedov, C.Yuan, 2026. A cosmic femtolensing Hawking radiation signature [URL](#), [DOI](#), (Cites: 0)
- [12] **Ntelis, P.**, 2026. [EinsteinKPy2Tex](#): A Python Tool for Computing and Exporting General Relativity Tensors to LaTeX. signature [URL](#), (Cites: 0)
- [13] **Ntelis, P.**, 2026. A temporal-spatial-entropic metric and its Kretschmann scalar [URL](#), (Cites: 0)

Outreach	2022- n_{ow} , Universions Youtube Channel
	2025 , Utube , Comet 3i ATLAS: Interstellar Halloween Secrets! (6')
	2025 , Utube , Are there Aliens ? Drake's Equation ! (7')
	2025 , Utube , Quantum Tunnelling in Your Palm! (1')
	2025 , Utube , Advanced Manifold-Metric Pairs (10')
	2025 , Utube , A sunny watery magic (33")
	2025 , Utube , Advanced Euler's identity (3')
	2025 , Utube , Analytical <i>poly</i> Λ CDM dynamics 101 (5' talk)
	2025 , Utube , Advancing Tensor Theories 101 (10' talk)
	2025 , Utube , Quintessence 101 (5' talk)
	2024 , Utube , Introduction cosmology for everyone (1h talk)
	2023('24), IBS of Provence , Neuroscience and cosmology for high school students (3h)
	2023 , IBS of Provence , Math and physics from CPT for high school students (3h)
	2022 , IBS of Provence , Mathematics and physics for high school students (3h)

Conferences	2026, HIT , Sino-Uzbek W, A cosmic whirlwind: New math and cosmic views, (45')
Talks/Posters	2026, HIT , SoP, Advanced math in cosmology (1h Seminar)
Seminars	2026, HIT , SoP, A note on Baye's data analysis (1h Seminar)
	2025, The Dark Side of the Universe , FAT dynamics to cosmology (25' Talk)
	2025, Cosmo From Home , Advanced math in cosmology (Talk)
	2024, 50 yr of Horndeski at PI , A probabilistic expanding universe under FA, (Poster)
	2024, Cosmo From Home , A probabilistic expanding universe under FA, (25 min talk)
	2023, NEB , New avenues of functors of actions theories, (Talk 20')
	2023, Cosmo From Home , New avenues on Functors of Actions , (25 min talk)
	2023, NTUA HET , New theories in cosmology and particle physics, (1h Seminar)
	2022, CERN TH DPT Visit , Functors of Actions (5' Blackboard Talk)
	2021, Euclid Science Talk , Overview of impact of interlopers on PNG (1h Talk)
	2020, Atelier D.E. , PNG with interlopers in Euclid, (20' Talk) HVT
	2020, Atelier D.E. , Perturbation Theory of LSS, code: CLASS-PT , (20' Talk)
	2020, SEMFE , NTUA , LSS: Theory and Observations, (40' Seminar)
	2019, 30th SYMPOSIUM , Fractality from Multiple Tracers , (15' Talk)
	2019, Euclid IAS Paris , Systematics on PNG, (20' Talk)
	2019, Euclid Consortium Helnsinki , Primordial Non-Gaussianity, (15' Talk)
	2019, eBOSS/DESI France , Cosmology with Cosmic Homogeneity (15' Talk)
	2019, Aix-University , Homogeneity with Multitracers , (2h Seminar)
	2019, Aix-University , A note on lss, Summer School , (6h Seminar)
	2018, Euclid GC-WL , Ameliorating cosmology with homogeneity, (15' Talk)
	2018, Colloquium of D.E. , Cosmic Homogeneity as a standard ruler, (20' Talk)
	2018, R. Moriond , <i>Euclid:Homogeneity in search of the Dark Sector</i> , (Poster) HIC
	2018, UiO , What can fractality tell us about the universe?, (1h seminar)
	2017, P.O.N.T. in Cosmology , <i>Is the universe homogeneous?</i> , (20' Talk)
	2016, R. Moriond , <i>The homogeneity scale of the universe</i> , (15' Talk)
	2015, UnivEarthS School , Santorini, <i>LSS and homogeneity</i> , (Poster)
	2015, Rencontres de Jeunes Physicienne , <i>Homogeneity (Video)</i> , (15' Talk)
	2014, Elbereth Conference , IAP , <i>The homogeneity of the universe</i> (15' Talk)
	2014, UnivEarthS School , <i>The homogeneity of the universe</i> , (3' Talk)

Software

- [1] P.Ntelis (2017-today): [Github contributions](#)
- [2] P.Ntelis (2017-today): [COSMOPIT](#)
- [3] P.Ntelis (2018-2021): [Gitlab private Euclid softwares](#)
- [4] P.Ntelis (2018): [MCMC-posterior-example](#)
- [5] P.Ntelis (2018): [CoHo](#)
- [6] P.Ntelis (2018): [ANALYSE-SN-magnitude-redshift-curve](#)
- [7] P.Ntelis (2018): [Extract-Cosmology-with-the-BA0-peak-position](#)
- [8] P.Ntelis (2020): [AofEFT](#)
- [9] P.Ntelis (2023): [Tutorial on galaxy simulations and coordinate transformation](#)
- [10] P.Ntelis (2025a): [Standard \$\phi\$ CDM dynamics](#)
- [11] P.Ntelis (2025b): [Simple quintessence dynamics](#)
- [12] P.Ntelis (2025c): [Analytical \$poly\Lambda\$ CDM dynamics](#)
- [13] P.Ntelis (2026): [EinsteinKPy2Tex](#)